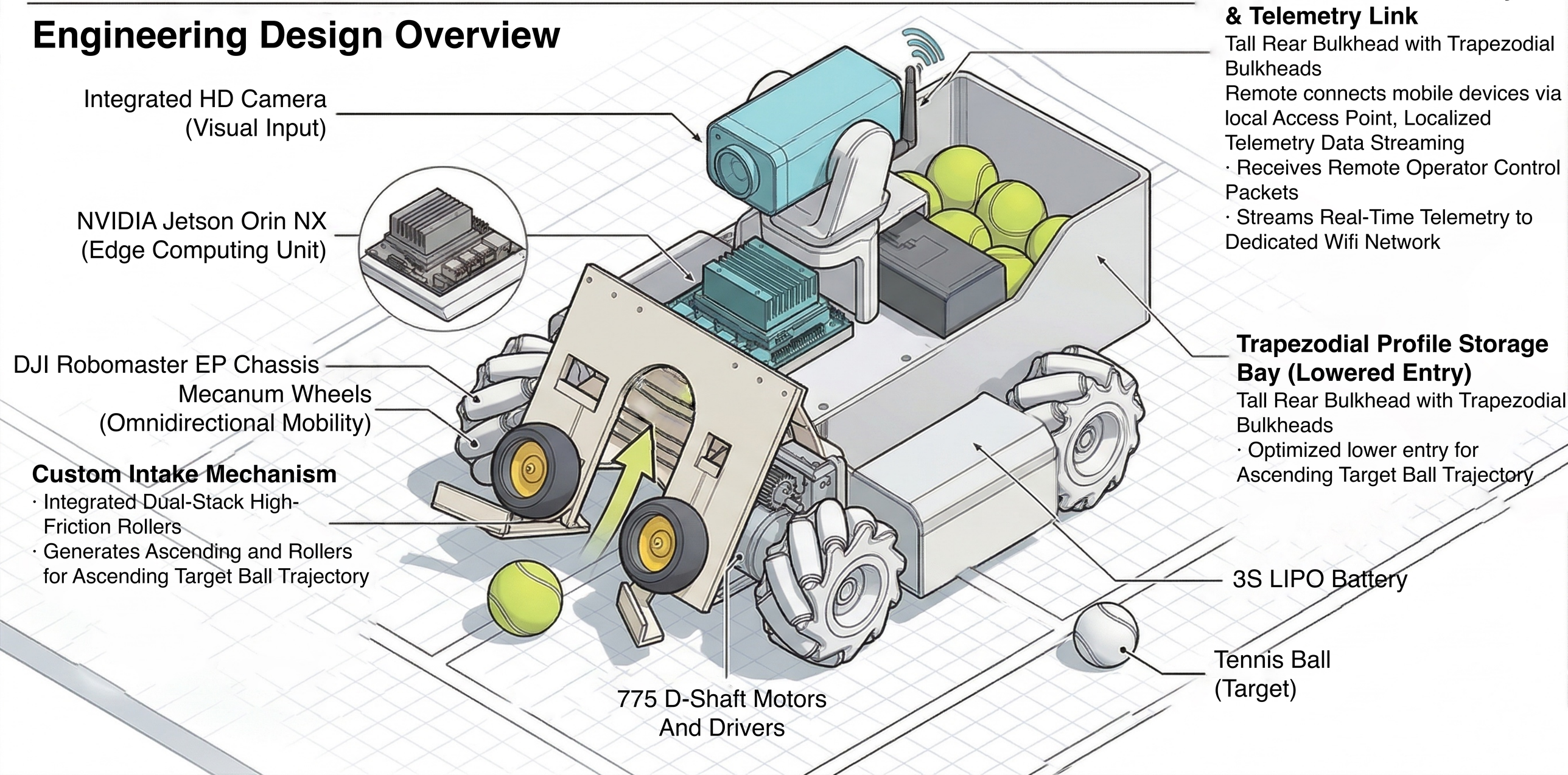
CourtSweeper: A Vision-based Tennis Ball Retriever

Engineering Design Overview



Remote Wifi Connectivity & Telemetry Link

Tall Rear Bulkhead with Trapezodial Bulkheads

Remote connects mobile devices via local Access Point, Localized Telemetry Data Streaming

· Receives Remote Operator Control Packets

· Streams Real-Time Telemetry to Dedicated Wifi Network

Trapezodial Profile Storage Bay (Lowered Entry)

Tall Rear Bulkhead with Trapezodial Bulkheads

· Optimized lower entry for Ascending Target Ball Trajectory

Custom Intake Mechanism

· Integrated Dual-Stack High-Friction Rollers

· Generates Ascending and Rollers for Ascending Target Ball Trajectory